



AWS SECURITY OPERATIONS CENTER

Phase 2 Validation & Cyber-Resiliency Stress Test

FINAL STATUS REPORT

February 22, 2026

CLASSIFICATION: CONFIDENTIAL — INTERNAL USE
ONLY

1. Executive Summary

This report documents the successful completion of Phase 2 validation for the AWS Cloud Security Operations Center (SOC). The SOC was built from the ground up using a dual-VPC architecture, Elastic Stack (SIEM), and multi-source log ingestion. All four validation categories were executed and passed.

Category	Objective	Result
A — Network & Communication	Cross-VPC peering, port isolation, agent heartbeat	PASS
B — Log Ingestion & Parsing	Auth logs, CloudTrail API mapping, RDS log fidelity	PASS
C — Cloud Native Pull Validation	S3 enumeration, VPC Flow REJECTs, metric accuracy	PASS
D — Adversarial Simulation	DDoS, malicious upload, SQL injection detection	PASS

2. Deployed Infrastructure

Component	Details
SOC Instance	t2.medium — Ubuntu 24.04 — Security VPC (10.1.1.25)
Elastic Stack	Elasticsearch + Kibana + Fleet Server v9.3.0 via Docker
Frontend EC2	ip-10-0-1-252 — Nginx Proxy — Main VPC Public Subnet
Backend EC2	ip-10-0-2-37 — Node.js API — Main VPC Private Subnet
RDS MySQL	company-db — Error + General logs enabled in CloudWatch
VPC Peering	Main VPC (10.0.0.0/16) <-> Security VPC (10.1.0.0/16)
IAM Role	SOC-Private-Instance-Role — Least privilege S3/CloudWatch/EC2

3. Active Log Sources

Source	Method	Index Pattern	Status
CloudTrail	S3 — aws-cloudtrail-logs-mj	logs-aws.cloudtrail-*	ACTIVE
VPC Flow Logs	CloudWatch — /aws/vpc/...	logs-aws.vpcflow-*	ACTIVE
RDS General Log	CloudWatch — /aws/rds/.../general	logs-aws.cloudwatch_logs-*	ACTIVE
RDS Error Log	CloudWatch — /aws/rds/.../error	logs-aws.cloudwatch_logs-*	ACTIVE

S3 Access Logs	S3 — company-storage-access-logs	logs-aws.s3access-*	ACTIVE
EC2 System Logs	Elastic Agent — local	logs-system.*	ACTIVE
EC2 Metrics	Elastic Agent — local	metrics-system.*	ACTIVE

4. Category A — Network & Communication

A.1 Cross-VPC Peering Latency

Validated sub-millisecond connectivity between Security VPC and Main VPC using TCP connection timing tests. The SOC instance successfully reached the Frontend EC2 and the Backend EC2 was confirmed reachable via the Frontend Security Group chain — consistent with the intended architecture.

Test Direction	Result	Status
Frontend -> SOC (9200)	0.000613s response time	PASS
SOC -> Frontend (80)	Reachable via peering	PASS
SOC -> Backend (80)	Blocked by SG design — correct	PASS

A.2 Port 9200/8220 Isolation

Verified that Elasticsearch (9200) and Fleet Server (8220) are inaccessible from public internet. Both ports demand authentication before returning any data. Security Group inbound rules confirmed to only allow traffic from VPC CIDRs — no 0.0.0.0/0 exposure on any critical port.

Port	Response	Status
9200 (Elasticsearch)	401 — Auth required before data returned	PASS
8220 (Fleet Server)	404 — Valid response, no data leak	PASS
5601 (Kibana)	Private subnet only — no public exposure	PASS

A.3 Agent Heartbeat Continuity

All three Elastic Agents (SOC, Frontend, Backend) confirmed healthy in Fleet with last activity timestamps within 30 seconds. Agents are configured to start on boot (systemd enabled) and auto-reconnect after interruption.

5. Category B — Log Ingestion & Parsing

B.1 EC2 Auth Log Correlation

Simulated SSH brute force attack using fake credentials against the Frontend EC2. Failed login events appeared in Kibana within 60 seconds under the logs-system.auth-* index, correctly populated with source IP, username attempted, and timestamp — fully ECS-mapped.

B.2 CloudTrail API Mapping

Executed IAM user creation and deletion to generate CloudTrail events. Both CreateUser and DeleteUser events appeared in Kibana under logs-aws.cloudtrail-* within 10 minutes, with correct aws.cloudtrail.user_identity.arn, event.action, and cloud.region fields populated.

B.3 RDS Log Fidelity

Tested both RDS log channels with targeted payloads:

- General Log: MySQL queries (SHOW DATABASES, SELECT SLEEP, UNION-based SQLi) captured in real time via CloudWatch polling.
- Error Log: Bulk connection flood (150 simultaneous connections) triggered MY-010055 reverse DNS warnings — confirmed in CloudWatch and subsequently ingested by Kibana.
- Key finding: Both log channels require dedicated CloudWatch integrations. Bundling multiple log groups in one integration caused silent polling failures.

6. Category C — Cloud Native Pull Validation

C.1 S3 Bucket Enumeration Detection

Executed 10 repeated aws s3 ls calls against company-storage-network-project from the SOC instance. ListBucket operations were captured in S3 access logs delivered to company-storage-access-logs bucket and ingested into Kibana under logs-aws.s3access-*.

C.2 VPC Flow Log REJECT Visibility

Generated blocked traffic by attempting connections to restricted ports on the Backend EC2 (22, 3306, 443) from the SOC instance. REJECT entries appeared in Kibana under logs-aws.vpcflow-* with correct source IP, destination port, and action fields — confirming Security Group enforcement is visible to the SIEM.

C.3 Metric Aggregation Accuracy

Deployed stress --cpu 4 on both Frontend and Backend EC2 instances simultaneously. CPU spike was visible in the Kibana System Host Overview dashboard under metrics-system.cpu-*. System integration metrics (CPU, memory, disk, network) confirmed active after enabling metric collection toggles in the Fleet agent policy.

7. Category D — Adversarial Simulation

D.1 Layer 7 DDoS Simulation

Flooded Frontend EC2 with 10,000+ HTTP requests using Apache Bench (ab) at 100 concurrent connections. Traffic surge was visible in VPC Flow Logs (logs-aws.vpcflow-*) as a flood of ACCEPT entries targeting the Frontend private IP. Network metrics confirmed spike in system.network dataset on the Kibana dashboard.

D.2 Malicious Payload Upload

Temporarily granted s3:PutObject to SOC role and uploaded a clearly-labelled test file (malware-test.txt) to company-storage-network-project. The PutObject event was captured in S3 access logs in Kibana. File was immediately deleted and the PutObject permission was revoked after the test. No production data was affected.

D.3 SQL Injection Probing

Executed UNION-based SQL injection payloads against the /api/files?search= endpoint via the Frontend proxy. Key findings:

- Kibana captured the raw SQLi payload in RDS General Log within 3 minutes: `SELECT ... WHERE filename LIKE '%\' UNION SELECT * FROM files--%'`
- MySQL escaping (backslash) prevented successful data exfiltration in most attempts.
- One unparameterized query path was identified — the SQLi payload appeared raw in the RDS general log confirming a real vulnerability exists.
- Remediation required: Parameterized queries must replace all raw string interpolation in the Node.js backend. AWS WAF deployment recommended as additional defense layer.

8. Security Findings & Remediation

#	Finding	Severity	Remediation	Status
1	SQL Injection vulnerability in /api/files endpoint	HIGH	Use parameterized queries + deploy AWS WAF	OPEN
2	Node.js v18 EOL — no longer receiving security updates	MEDIUM	Upgrade to Node.js v20 LTS or v22 LTS	OPEN
3	RDS reverse DNS resolution warnings under load	LOW	Enable DNS hostnames in VPC settings for RDS subnet	MONITOR
4	CloudWatch polling delay (3-5 min) for /error logs	INFO	Acceptable for lab — tune poll interval in production	ACCEPTED

9. SOC Graduation Sign-Off

The AWS Cloud SOC has been validated against all Phase 2 requirements. All four testing categories passed their success conditions. The SIEM is receiving, parsing, and surfacing actionable security telemetry from all defined log sources.

Metric	Result
Total Test Categories	4 of 4 PASSED
Total Log Sources Active	8 of 8 ACTIVE
Agents Enrolled & Healthy	3 (SOC + Frontend + Backend)
Critical Ports Publicly Exposed	0
Attacks Detected by SIEM	All simulated attacks captured
High Severity Findings	1 (SQLi — remediation required)
SOC Operational Status	BATTLE-READY

Phase 2 Complete — SOC Validated & Operational

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